T, TD - Solving trig equations

Theory

Using trig identities, it is easy to solve trig equations like these:

$$(1) \qquad 4\cos^2\theta - 4\sin\theta = 1$$

$$(2) \qquad \cos(2\theta) + \sin\theta = 0$$

Convert to polynomial

Notice in (1) the mismatch of types: $\cos\theta$ in the square but $\sin\theta$ without the square.

Notice in (2) the mismatch of inputs: 2θ in the cosine and θ in the sine.

Method: In either situation, convert to a single *type* and a single *input* using trig identities.

The result has the form of polynomial equation that can be solved.

Examples

Example (1):

$$\begin{aligned} 4\cos^2\theta - 4\sin\theta = 1 \quad &\ggg \quad 4(1-\sin^2\theta) - 4\sin\theta = 1 \\ &\overset{X=\sin\theta}{\ggg} \quad 4(1-X^2) - 4X = 1 \\ &\ggg \quad -4X^2 - 4X + 3 = 0 \\ &\ggg \quad X = -\frac{3}{2}, \frac{1}{2} \\ &\ggg \quad \theta = \frac{\pi}{6}, \frac{5\pi}{6} \end{aligned}$$

Example (2):

$$
\begin{aligned}
\cos(2\theta) + \sin\theta = 0 \quad & \ggg \quad \cos^2\theta - \sin^2\theta + \sin\theta = 0 \\
& \ggg \quad 1 - 2\sin^2\theta + \sin\theta = 0 \\
& \ggg \quad -2X^2 + X + 1 = 0 \\
& \ggg \quad X = -\frac{1}{2},\, 1 \\
& \ggg \quad \theta = \frac{\pi}{2},\, \frac{7\pi}{6},\, \frac{11\pi}{6}
\end{aligned}
$$

Problems

- Solve these equations and verify your answers.
- If your answer is wrong, try to find your mistake and fix your solution.
- Afterwards, study the full solution provided in a separate file.

Assume that x and θ lie in $[0,\, 2\pi)$ for all these problems.

1. $$\cos(2x) - 2\cos x + 1 = 0$$

Answer: $x = 0,\ \frac{\pi}{2},\ \frac{3\pi}{2}$

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2. $$\sin(2x) + \sin x = 0$$

Answer: $x = 0,\ \pi,\ \frac{2\pi}{3},\ \frac{4\pi}{3}$

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3. $$\cos(2x) + \cos x + 1 = 0$$

Answer: $x = \frac{\pi}{2},\ \frac{3\pi}{2},\ \frac{2\pi}{3},\ \frac{4\pi}{3}$

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4. $$\cos(2x) + \frac{1}{2} = 0$$

Answer: $x = \frac{\pi}{3},\ \frac{2\pi}{3},\ \frac{4\pi}{3},\ \frac{5\pi}{3}$

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5. $$\tan(2\theta) - 3\tan\theta = 0$$

Answer: $\theta = 0,\ \frac{\pi}{6},\ \frac{5\pi}{6},\ \pi,\ \frac{7\pi}{6},\ \frac{11\pi}{6}$

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6. $$\sin^2\theta + \sin\theta - 2 = 0$$

Answer: $\theta = \frac{\pi}{2}$

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7. $$\sin(2\theta) = \cos\theta$$

Answer: $\theta = \frac{\pi}{6},\ \frac{\pi}{2},\ \frac{5\pi}{6},\ \frac{3\pi}{2}$

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8. $$\cos(2\theta) - \sin\theta = 0$$

Answer: $\theta = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{3\pi}{2}$

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9. $$3\sec^2\theta - 2\tan^2\theta - 4 = 0$$

Answer: $\theta = \frac{\pi}{4},\ \frac{3\pi}{4},\ \frac{5\pi}{4},\ \frac{7\pi}{4}$